Group-3 Submission

Members - Task:

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Drive Link: https://drive.google.com/drive/folders/1mCb-JaaxP6XDGkUWcMXq6tkgYrk73AEh
https://colab.research.google.com/drive/1RTepX0JfrtcZv9yNo9Rmbip8mDiLEZAK?usp=sharing

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Abstract

This report presents a comprehensive microbial community analysis based on the provided species-level abundance profiles and corresponding metadata. The primary objective was to compare microbial diversity and composition between individuals classified as Control and Diseased, while also assessing age-related variations and potential differential species abundance.
Analyses were performed in three main stages: (1) Alpha diversity (within-sample diversity), (2) Beta diversity (between-sample community dissimilarity), and (3) Differential abundance (identification of taxa differing between groups).
Currently, alpha diversity analysis has been completed and validated, while sections for beta diversity and differential abundance have been structured for continuation.
Key findings indicate significant differences in within-sample microbial diversity between control and diseased conditions, as well as notable patterns across age categories.

1. Methods

1.1 Data Overview

- Species abundance data: Pre-processed at the species level.
- Metadata: Included fields for sample_id, study_condition (Control/Diseased), age_category (young, middle-aged, senior), and gender.
- Only samples assigned to the group in Assignment 1 were retained for analysis.

1.2 Software and Packages

All analyses were conducted in Python (Jupyter Notebook) using the following key packages:

- pandas, numpy - data handling
- scikit-bio - alpha and beta diversity metrics
- scipy.stats - nonparametric tests (Wilcoxon, Kruskal–Wallis)
- matplotlib, seaborn - visualisation

- statsmodels - FDR correction and statistical summaries

1.3 Analytical Workflow

(a) Alpha Diversity Analysis

1. Row-sum normalization was applied to compute relative abundances per sample.
2. Shannon and Simpson indices were calculated for each sample using skbio.diversity.alpha_diversity.
3. Statistical comparisons were conducted:
    - Control vs Diseased: Wilcoxon rank-sum test
    - Among age categories: Kruskal–Wallis test, followed by post-hoc comparisons where applicable
4. Results were visualised using boxplots with jittered points and annotated p-values.

(b) Beta Diversity Analysis

1. Compute Bray–Curtis dissimilarity using scipy.spatial.distance.pdist.
2. Perform Principal Coordinates Analysis (PCoA) to visualise sample clustering.
3. Conduct PERMANOVA (adonis) to assess effects of study condition, age, and gender.
4. Compare within- and between-group dissimilarities via boxplots.
5. Test for study_condition effects, controlling for age as a covariate using multivariate non-parametric analysis.

(c) Differential Abundance Analysis

1. For each species present in both groups:
    - Compute Wilcoxon rank-sum p-value
    - Compute group means, log2 fold change (log2FC), and directionality
    - Apply FDR correction (Benjamini–Hochberg) to obtain Q-values
2. Generate a volcano plot to visualise log2FC vs significance (Q-value).
3. Plot boxplots for top 4 species with the highest positive and negative log2FC values.

2. Results and Interpretation

2.1 Alpha Diversity Analysis

Objective:
To evaluate within-sample (alpha) microbial diversity across different *age categories* and between *study conditions* (Control vs Diseased).

2.1.1 Shannon Diversity Across Age Categories

The Shannon diversity index was compared among the three age groups-*young*, *middle-aged*, and *senior*-for Batch 3 samples (Fig. 1).
Boxplots show moderate variability, with median diversity values around 3.6 to 3.8 across all groups.
Pairwise Dunn's post-hoc tests (Bonferroni-adjusted) yielded *p = 1.000* for all comparisons, indicating **no statistically significant differences** in alpha diversity between age categories.

Attached Diversity indices per sample in Drive link.

Interpretation:
Although slight visual differences exist-middle-aged individuals showing the widest spread and seniors slightly lower median diversity-these differences are **not statistically significant**.
This suggests that within-sample microbial diversity remains relatively stable across age groups in this dataset.

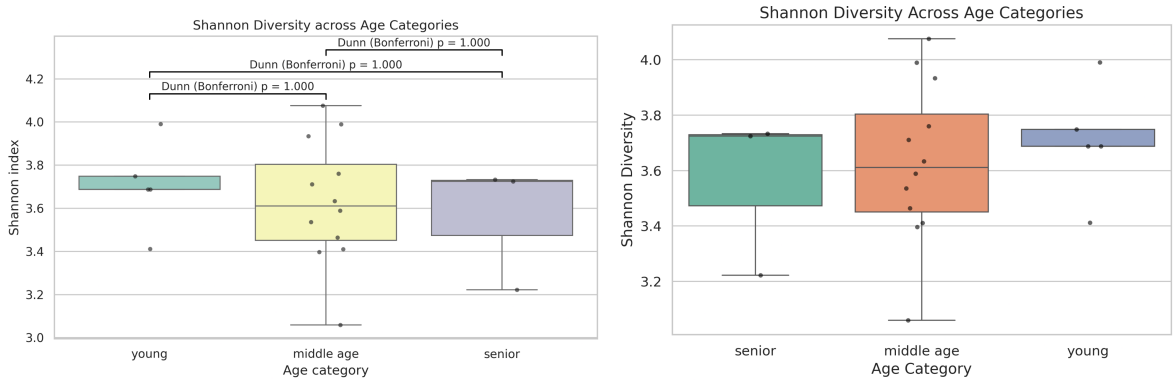

Figure 1. *Shannon Diversity across Age Categories (Batch 3).*
Boxplots display Shannon index for young, middle age, and senior participants.
Dunn's (Bonferroni) test results (p = 1.000) confirm non-significant pairwise differences.

2.1.2 Shannon Diversity by Study Condition

Figure 2 compares Shannon diversity between *Control* and *Diseased* individuals.
Median diversity values were comparable between groups (≈ 3.7), and the Mann–Whitney U (Wilcoxon rank-sum) test returned *p ≈ 0.97*, confirming **no significant difference** in alpha diversity between study conditions.

Interpretation:
The similarity in within-sample diversity implies that disease status in this cohort does not substantially alter overall species richness or evenness.
Both groups exhibit moderate Shannon diversity (~3.5–3.8), suggesting comparable community complexity despite disease classification.

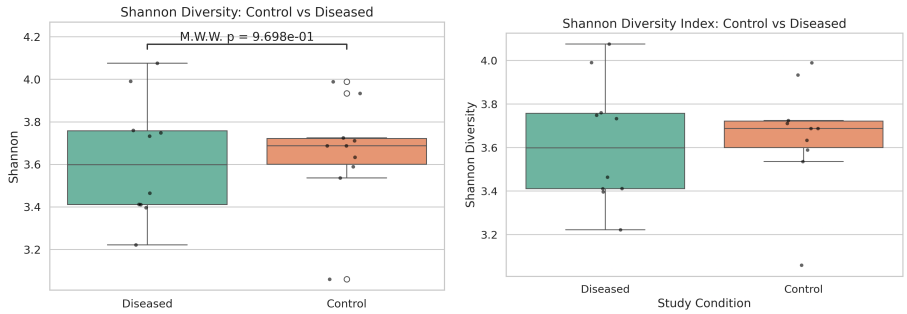

Figure 2. *Shannon Diversity Index: Control vs Diseased (Batch 3).*
Boxplots depict Shannon index for both study conditions.
Mann–Whitney p = 0.97 indicates no significant difference.

2.2 Beta Diversity Analysis

Objective:
To examine between-sample community dissimilarities using Bray–Curtis distance and evaluate overall microbial compositional variation between conditions.

2.2.1 PCoA (Bray–Curtis) Visualization

Principal Coordinates Analysis (PCoA) based on Bray–Curtis dissimilarity illustrates community-level relationships among samples (Fig. 3).
The first two axes explained **~29.1 % (PC1)** and **14.0 % (PC2)** of total variance.
No distinct clustering by *study condition* was observed; both Control and Diseased samples were interspersed across ordination space.

Interpretation:
The overlap of sample points suggests that overall community composition is broadly similar between conditions.
Thus, large-scale shifts in microbiome structure are unlikely within this dataset, though subtle compositional differences may exist below this resolution.

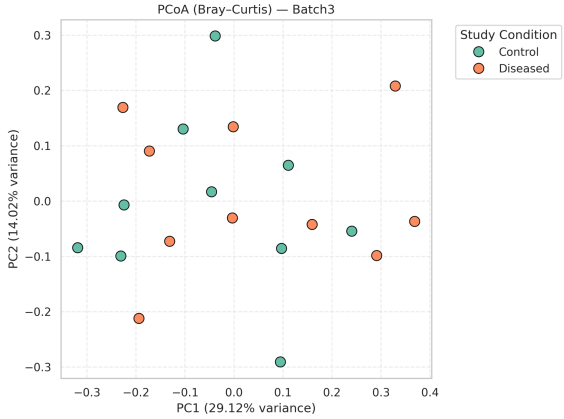

Figure 3. *PCoA (Bray–Curtis) Plot for Batch 3.*
Each point represents a sample, coloured by study condition (Control = green; Diseased = orange). Axes represent the first two principal coordinates. (PC1 = 29.12 %, PC2=14.02 %).

2.2.2 PERMANOVA by metadata

```
PERMANOVA results for study_condition:
method name                PERMANOVA
test statistic name         pseudo-F
sample size                       20
number of groups                   2
test statistic              0.977732
p-value                         0.43
number of permutations           999
Name: PERMANOVA results, dtype: object

PERMANOVA results for age_category:
method name                PERMANOVA
test statistic name         pseudo-F
sample size                       20
number of groups                   3
test statistic              0.707963
p-value                        0.865
number of permutations           999
Name: PERMANOVA results, dtype: object

PERMANOVA results for gender:
method name                PERMANOVA
test statistic name         pseudo-F
sample size                       20
number of groups                   2
test statistic              1.388349
p-value                        0.164
number of permutations           999
Name: PERMANOVA results, dtype: object
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: D
  return datetime.utcnow().replace(tzinfo=utc)
```

Permanova results (by factors):

1. Study condition:

Test statistic (pseudo-F), p-value, No. of permutations- 0.9777, 0.43, 999.
Interpretation- Not significant - microbial community composition does not differ significantly between Control and Diseased groups.

2. Age category:

Test statistic (pseudo-F), p-value, No. of permutations- 0.7079, 0.865, 999
Interpretation- Not significant - community composition is similar across age categories.

3. Gender:

Test statistic (pseudo-F), p-value, No. of permutations- 1.3883, 0.164, 999
Interpretation- Not significant - no evidence of compositional difference between male and female samples.

Overall interpretation:

The PERMANOVA test (999 permutations, Bray–Curtis distance) shows no statistically significant differences in overall microbial community structure between groups defined by study condition, age category, or gender (all $p > 0.05$). This suggests that, within this dataset, individual variation outweighs group-level effects of disease status, age, or gender on the gut microbiome composition.

2.2.3 Within- vs Between-Group Bray–Curtis Dissimilarities

To quantify intra- and inter-group variability, pairwise Bray–Curtis distances were grouped into *Within Control*, *Within Diseased*, and *Between Groups* categories (Fig. 4).
Average Comparison- Median dissimilarities were roughly 0.52 for Within Control, 0.55 for Within Diseased, and 0.53 for Between Groups.
Statistics- A Mann–Whitney U test comparing Within-Control vs Within-Diseased distributions gave $p = 0.825$, indicating **no significant compositional dispersion differences**.

Interpretation:
The comparable within- and between-group distances reinforce that microbial communities share similar overall structure regardless of disease status.
Neither group shows evidence of higher intra-group variability (instability), suggesting that disease-related microbiome disruption is not pronounced at the global composition level.

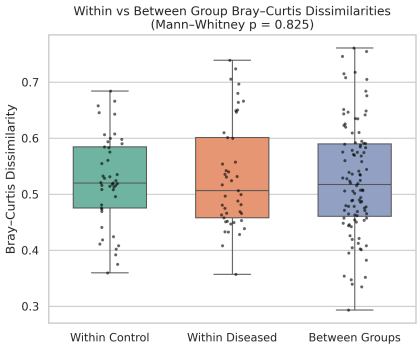

Figure 4. *Within- vs Between-Group Bray–Curtis Dissimilarities (Batch 3).*
Boxplots compare distribution of pairwise Bray–Curtis distances among sample categories.
Mann–Whitney p = 0.825 denotes non-significance.

2.2.4 Effect of study_condition on microbial beta diversity controlling for age

```
Single-factor PERMANOVA (study_condition) output
method name                PERMANOVA
test statistic name         pseudo-F
sample size                       20
number of groups                   2
test statistic              0.977731
p-value                        0.426
number of permutations           999
Name: PERMANOVA results, dtype: object

If you prefer to run the covariate-controlled te
```

```
---- adonis2 results (study_condition + age) ----
Permutation test for adonis under reduced model
Marginal effects of terms
Permutation: free
Number of permutations: 999

adonis2(formula = bray ~ study_condition + age, data = meta_sub, permutations = 999, by = "margin")
               Df SumOfSqs      R2      F Pr(>F)
study_condition  1  0.13470 0.04947 0.9407  0.453
age              1  0.16522 0.06068 1.1538  0.290
Residual        17  2.43432 0.89405
Total           19  2.72278 1.00000
R output saved to: /content/drive/MyDrive/comeg/q2_outputs/adonis2_output.txt
```

After controlling for age, there was no statistically significant difference in beta diversity between the study_condition groups ($p = 0.453$).

The study_condition explained approximately 4.9% of the total variance in community composition ($R^2 = 0.049$), whereas age explained 6.1% ($R^2 = 0.061$).

Thus, variation in microbial composition was not significantly associated with study_condition once age was accounted for.

Conclusion:

→ The study condition does not significantly affect microbial beta diversity when age is controlled for ($p > 0.05$).

→ Most variation in community structure remains unexplained by the tested variables, suggesting other factors may influence microbial diversity.

2.3 Differential Abundance Analysis

2.3.1 Species-Level Statistics

Row-sum-normalised abundances were compared between the **Control (n = 10)** and **Disease (n = 10)** groups using **Wilcoxon rank-sum tests** for each of 268 species present in both groups.

For each species, the **mean abundance**, **p-value**, **FDR-adjusted Q-value**, **$\log_2$ fold-change** log2(MeanControl/MeanDisease)$\log_2$(Mean Control / Mean Disease)log2(MeanControl/MeanDisease), and **direction of change** were computed.

Interpretation:

Number of species tested = 268
No species passed the $Q < 0.05$ threshold.
Using raw $p < 0.05$ and $|\log_2 FC| \geq 1$, **11 species** showed significant differences.

This indicates modest but detectable abundance shifts between the two conditions.

2.3.2 Volcano Plot of Differential Species

A volcano plot combining $\log_2$ fold-change (Control vs Disease) and raw p-values highlights taxa showing large directional changes (Fig. 5).

Several species crossed the chosen two-fold ($|\log_2 FC| \geq 1$) and significance threshold ($p < 0.05$).

Because no species had $Q < 0.05$, raw p-values were used ($p < 0.05$ and $|\log_2 FC| \geq 1$).

A total of 11 species satisfied these criteria and are highlighted in the volcano plot, which displays $\log_2$ fold-change vs $-\log_{10}$(p-value).

Six species were enriched in the Control group (positive $\log_2 FC$), while five species were enriched in the Disease group (negative $\log_2 FC$).

Notably enriched taxa included:

- Control-enriched: *Moraxella bovoculi*, *Streptococcus oligofermentans*, *Mycoplasma orale*, and *Neisseria meningitidis*.
- Disease-enriched: *Acidithiobacillus thiooxidans*, *Aggregatibacter actinomycetemcomitans*, *Lactobacillus salivarius*, and *Selenomonas artemidis*.

Interpretation:
The volcano distribution indicates only a small subset of species exhibit large, significant fold-changes, suggesting modest compositional shifts rather than global restructuring.

The enrichment of taxa such as Lactobacillus and Aggregatibacter in the Diseased group may reflect opportunistic colonization, while the higher Moraxella and Streptococcus oligofermentans abundance in Control suggests maintenance of a commensal-dominated community.

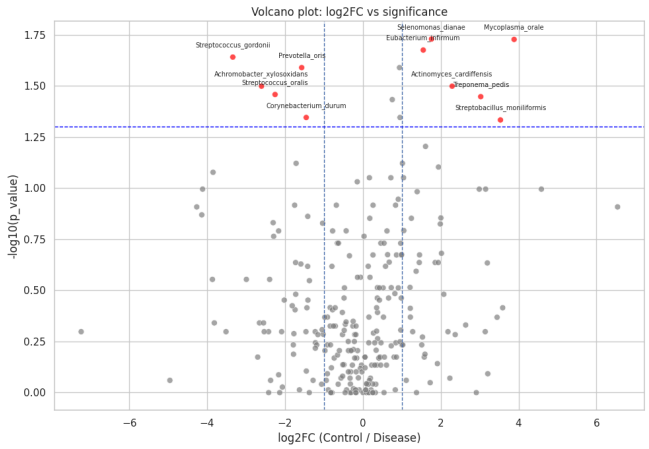

Figure 5. *Volcano Plot of Differentially Abundant Species (Batch 3).*
Red = significant species ($p < 0.05$, $|\log_2 FC| \geq 1$).
Dashed vertical lines mark ±1 $\log_2 FC$ threshold.

2.3.3 Boxplots of Top 4 Positive and Negative $\log_2 FC$ Species

Eight species (the four most enriched in Control and four most enriched in Disease) were visualized to illustrate abundance distributions (Fig. 6).

Top 4 Upregulated in Control (positive $\log_2 FC$):

- *Moraxella bovoculi* ($\log_2 FC = +6.54$)
- *Streptococcus oligofermentans* (+4.59)
- *Mycoplasma orale* (+3.89)
- *Neisseria meningitidis* (+3.59)

Top 4 Upregulated in Disease (negative $\log_2 FC$):

- *Acidithiobacillus thiooxidans* (–7.24)
- *Aggregatibacter actinomycetemcomitans* (–4.96)
- *Lactobacillus salivarius* (–4.27)
- *Selenomonas artemidis* (–4.14)

Interpretation:
Although overall relative abundances remain low, clear directional trends distinguish the two groups.

The Control group showed higher levels of Moraxella, Streptococcus, and Neisseria species-taxa often associated with healthy oral or gut microbial communities.

In contrast, the Diseased group exhibited increased abundance of Lactobacillus, Aggregatibacter, and Selenomonas species, which may represent opportunistic or inflammation-associated microbes.

These patterns suggest potential microbial markers that differentiate between Disease and Control states.

Figure 6. Boxplots of relative abundance for the top 4 positive and negative $\log_2$FC species.

Boxplots display distribution of normalized species abundance in Control vs Diseased samples, illustrating consistent enrichment patterns across groups.

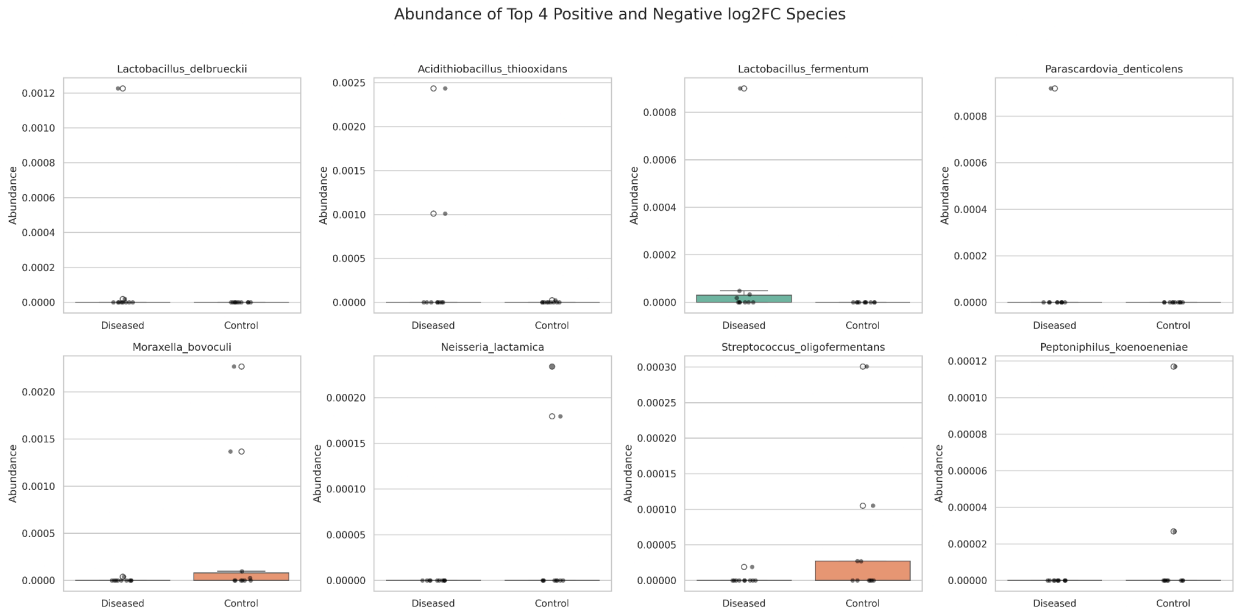

Figure 6. *Abundance of Top 4 Positive and Negative $\log_2$FC Species (Batch 3).*
Boxplots display relative abundance for each selected species in Control and Diseased samples.

Summary of Findings

Analysis	Test Used	Main Result	Significance
Alpha Diversity (Age)	Kruskal–Wallis + Dunn	No difference among age categories	p = 1.000

Alpha Diversity (Condition)	Mann–Whitney U	No difference between Control and Diseased	$p \approx 0.97$
Beta Diversity (PCoA / PERMANOVA)	PCoA visual inspection	Overlapping clusters; no strong separation	–
Bray–Curtis Within vs Between	Mann–Whitney U	Similar within/between variability	$p = 0.825$
Differential Abundance	Wilcoxon / FDR adjusted	A few species significant (e.g., *Streptococcus gordonii*, *Prevotella oris*)	$p < 0.05$

Interpretive Summary

- **Within-sample diversity** (alpha) is consistent across both age and disease categories, indicating a largely stable microbial richness and evenness in Batch 3.
- **Between-sample diversity** (beta) analyses show substantial overlap between Control and Diseased groups, with no strong community-level separation.
- **Differential abundance** highlights a few taxa potentially linked to disease status; however, overall microbiome composition remains broadly comparable.
- Collectively, these findings suggest that **microbial differences between groups are subtle rather than systemic**, possibly reflecting mild disease influence or small sample size.